

Non-ergodic Delocalized States and Quantum Algorithms

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Introduction

This text in quantum theory is not original but a summary and explanation of ideas proposed in various relevant papers in the field. The first part is mostly a summary of the important theoretical concepts which are fundamental to the discussion of (multi)fractal states and their interesting properties. Whilst the second part aims to investigate more detail as to how they might prove to be particularly useful (or not) for speeding up quantum algorithms in certain circumstances. The third part is a summary of other novel methods and discussions about quantum algorithms and mechanisms. Please feel free to catch any error or misunderstanding if there is any.

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1 Theoretical Concepts

We formulate some basic terminology and theory in this section, starting from the time-independent Schrödinger equation (TISE)

$$H |\psi_E\rangle = E |\psi_E\rangle \quad (1)$$

where H is the Hamiltonian of a standard quantum system with disorder involved, and $|\psi_E\rangle$ is an eigenstate of the system with associated eigenenergy E . The wavefunction ψ_E can be specifically characterized (in position basis) as

$$\psi_E(n) = \langle n | \psi_E \rangle \quad (2)$$

where $n = (1, 2, \dots, N)$ is the site label in a sample of N total sites, with the normalization condition given by

$$\sum_{n=1}^N |\psi_E(n)|^2 = 1 \quad (3)$$

1.1 Fractal Dimensions

To distinguish states, one introduces the generalized inverse participation ratio, concerning how the q -th moment of a probability density scales with system size N :

$$P_q = \sum_{n=1}^N |\psi_E(n)|^{2q} \sim N^{(1-q)D_q} \quad (4)$$

where D_q specifies the generalized **fractal dimension** of order q in our disordered system [1]. Specifically one notes that if $D_q = 0$, then the system is **localized** to exactly one site $n = n_0$ which has $|\psi_E(n = n_0)|^2 = 1$, whilst all the other $|\psi_E(n \neq n_0)|^2 = 0$. At the other end, if $D_q = 1$, then the system is **fully ergodic**, as all $|\psi_E(n)|^2 = 1/N$ independent of site n , leading to $P_q = N(1/N)^q = N^{1-q}$ as naturally expected. For the intermediate case where $0 < D_q < 1$, we arrive at the **non-ergodic extended (NEE)** phases of matter, involving fractal (or multifractal) states.

Indeed it is possible to approximately calculate the moment by taking the limit $N \rightarrow \infty$ to convert to an integral [2]. To do this, one can introduce a (local) scaling exponent α defined as

$$|\psi|^2 \simeq N^{-\alpha} \quad (5)$$

or equivalently

$$\alpha \simeq -\frac{\ln |\psi|^2}{\ln N}$$

To count the number of sites with an associated exponent α , the multifractal spectrum function $f(\alpha)$ is defined as follows

$$\#(|\psi|^2 \simeq N^{-\alpha}) \sim N^{f(\alpha)} \quad (6)$$

Considering a system of total N sites, we naturally derive the probability distribution integrand

$$P(|\psi|^2 = N^{-\alpha}) d|\psi|^2 \sim N^{f(\alpha)-1} d\alpha. \quad (7)$$

Therefore, using (6) and (7), we get the probability normalization condition

$$1 = \int P(|\psi|^2) d|\psi|^2 \sim \int N^{f(\alpha)-1} d\alpha \sim N^{\max_{\alpha} f(\alpha)-1} \quad (8)$$

where last approximation follows from the fact that large exponent dominates the most.

And we also have

$$P_q = \sum_n |\psi(n)|^{2q} = \int N |\psi|^{2q} P(|\psi|^2) d|\psi|^2 \sim \int N^{f(\alpha)-q\alpha} d\alpha \sim N^{\max_{\alpha} [f(\alpha)-q\alpha]} \quad (9)$$

where setting $q = 1$, $P_q = 1$ will retrieve the normalization condition for ψ as a special case. This quantitative calculation allows us to connect the exponent α with the **eigenstate fractal dimension** D_q [2]. By $P_q \sim N^{\max_{\alpha} [f(\alpha)-q\alpha]}$ and $P_q \sim N^{(1-q)D_q}$, one naturally has

$$(1-q)D_q = \max_{\alpha} [f(\alpha) - q\alpha] \quad (10)$$

which also implies that

$$\frac{d}{d\alpha} (f - q\alpha) = 0$$

and

$$q = \frac{df}{d\alpha} \quad (11)$$

Analysing at the maximum, then we obtain $f(\alpha)$ in the form of a Legendre transform as follows

$$f(\alpha) = q\alpha + (1-q)D_q \quad (12)$$

which relates a local exponent α with the global scaling dimension D_q .

One can also show that the maximum value of $f(\alpha_0) = \max_{\alpha} f(\alpha)$ is 1 to satisfy normalization conditions, and for $N \rightarrow \infty$, the number $N^{-\alpha_0}$ will prevail to be the *typical* value of the ensemble of states [3].

1.2 Classification of States

From the perspective of quantum state dynamics, one can physically also distinguish these phases using the following limits involving the Hamiltonian matrix elements H_{ij} [4].

(I) **Anderson localization.**

$$\lim_{N \rightarrow \infty} \sum_j \langle |H_{ij}| \rangle < \infty \quad (13)$$

which means that a given site has only a finite number of resonant neighbors, the wavefunction cannot spread indefinitely. Quantum mechanically it is the convergence of perturbation theory in resonance counting arguments.

(II) **Mott's ergodicity principle.**

$$\lim_{N \rightarrow \infty} \sum_j \langle |H_{ij}|^2 \rangle \rightarrow \infty \quad (14)$$

which is derived from **Fermi's Golden Rule** $\Gamma_i = 2\pi\rho \sum_j |H_{ij}|^2$. Hence, in this ergodic case the escape (transition) rate Γ is much larger than the disorder bandwidth, and the particle spreads out.

(III) **NEE (Fractality).**

$$\sum_j \langle |H_{ij}| \rangle \rightarrow \infty \quad (15)$$

and

$$\sum_j \langle |H_{ij}|^2 \rangle \rightarrow 0 \quad (16)$$

where the Anderson localization criterion is violated, but the Fermi Golden Rule criterion for ergodicity is not satisfied. Here the states are sitting at the intermediate region of extended (delocalized) but non-ergodic phases.

1.3 Rosenzweig-Porter Model and Fractality

We here analyze the **Rosenzweig-Porter model** which provides a good theoretical basis for ergodic, localized, and fractal phases by adding matrix elements with disorder. Although there are claims for it to host **multifractal** states, it is calculated to be just **fractal**, namely that D_q is constant for integer positive q [3].

1.3.1 Gaussian Rosenzweig-Porter model

We first look at the **Gaussian Rosenzweig-Porter model**, which has entries of $N \times N$ Gaussian random matrices of the form

$$H_{RP} = H_0 + V \quad (17)$$

where $[H_0]_{ij} = \epsilon_i \delta_{ij}$, $\epsilon_u \in \mathcal{N}(0, 1)$ and $V = N^{-\gamma/2} H_{GOE}$, with H_{GOE} being a Gaussian Orthogonal Ensemble of elements with zero mean and unit variance. As will be shown, γ characterizes the strength of the off-diagonal hopping relative to the diagonal disorder, setting key boundaries for Anderson localization and ergodic transition, etc.

To calculate self-consistently the spectrum of fractal dimensions and the factor D_q we can do it via the cavity method, by writing down the Green's function

$$G(z) = (z - H)^{-1}$$

where $z = \epsilon - i\eta$, $H = H_{RP}$. Via the Schur complement one can show that the ii element of $G(z)$

is given by

$$G_{ii}^{-1} = z - \epsilon_i - V_i^\top \left(z - H^{(i)} \right)^{-1} V_i$$

where $H^{(i)}$ and H differ by a single site i which is the "cavity", so one can frame the reduced Green's function $G^{(i)}(z) = (z - H^{(i)})^{-1}$, leading to the result

$$G_{ii} = \left(z - \epsilon_i - \sum_{j,k \neq i} V_{ij} G_{jk}^{(i)} V_{ki} \right)^{-1} \simeq \left(z - \epsilon_i - \sum_{j \neq i} |V_{ij}|^2 G_{jj}^{(i)} \right)^{-1} \quad (18)$$

where the last equality is possible as we make use of the diagonal approximation where the diagonal elements of $G^{(i)}(z)$ dominate in the thermodynamic limit $N \rightarrow \infty$, while the system is also in a non-ergodic phase.

To proceed in calculation, one can define a local density of states (LDOS)

$$\nu_i(\epsilon) = \frac{1}{\pi} \text{Im}\{G_{ii}\} \quad (19)$$

which will turn out to be in a Lorentzian shape with respect to the eigenenergy variable ϵ . For simplicity, define the diagonal self-energy as

$$\Sigma_i = \Delta_i - i\Gamma_i = \sum_{j \neq i} |V_{ij}|^2 G_{jj}^{(i)}$$

and we have

$$G_{ii} = \frac{1}{(\epsilon - \epsilon_i - \Delta_i) - i(\eta + \Gamma_i)}$$

which leads to

$$\nu_i(\epsilon) \sim \frac{\eta + \Gamma_i}{(\epsilon - \epsilon_i - \Delta_i)^2 + (\eta + \Gamma_i)^2}$$

Here if one assumes that the real part of the diagonal self-energy compared to the on-site disorder amplitude ($\Sigma_i \ll \epsilon_i$) and that the parameter η is negligible compared to the broadening Γ_i ($\eta \ll \Gamma_i$). It should be emphasized that these approximations hold in the NEE phase only. Then, we obtain the formula for the LDOS:

$$\nu_i \sim \frac{\Gamma_i}{(\epsilon - \epsilon_i)^2 + \Gamma_i} \quad (20)$$

where, being the imaginary part of the diagonal self-energy, there exists the relation

$$\Gamma_i(\epsilon) = \sum_{j \neq i} |V_{ij}|^2 \nu_j(\epsilon) \quad (21)$$

Through graphical methods to solve the self-consistent problem, one can show that $\Gamma_i \sim N^{-c}$ is the typical value parametrized by $\alpha_0(\Gamma_i) = c$. A graphical method for calculating composite random variables specified by (20) also gives $\sum_{j \neq i} |V_{ij}|^2 \nu_j(\epsilon) \sim N^{\gamma-1}$ [3]. Hence, for the Gaussian

Rosenzweig-Porter model, we get (in the NEE phase)

$$c = \gamma - 1 \quad (22)$$

The case $c = 0, \gamma_{ET} = 1$ gives the ergodic limit ($\Gamma \sim N^0$, all roughly of the same magnitude for elements ν_i). The Anderson transition limit is at $c = 1, \gamma_{AT} = 2$ where $\Gamma \sim N^{-1}$, and a fraction of large components mainly contributes.

To associate the fractal dimension D_q with the parameter γ , consider the number of states M inside one resonance, given by $M \sim \Gamma/\delta_\epsilon \sim N^{-c}/N^{-1} \sim N^{1-c}$, where Γ is the broadening width, and δ_ϵ is the measure of a mean level spacing in the spectrum. Thus, we have the inverse participation ratio $P_q \sim M(1/M)^q \sim N^{(1-c)(1-q)}$. Compare the form to (4), we get

$$D_q = 1 - c = 2 - \gamma \quad (23)$$

which is expected, as $\gamma = 1, D_q = 1$ and $\gamma = 2, D_q = 0$ correspond to the ergodic and localized limits respectively. It is simply fractal (not multifractal) since there is no dependence on q for D_q .

1.3.2 Lévy-Rosenzweig-Porter model

The **Lévy-Rosenzweig-Porter model** differs from the Gaussian model in the distribution by involving fat-tailed Lévy amplitudes. The off-diagonal elements no longer follow a Gaussian ensemble but are distributed according to the following PDF with parameters γ and μ : (this is not the only parametrization, and here the convention is the same as in [3])

$$p(V_{ij}) \sim \frac{2\mu N^{-\mu\gamma/2}}{|V_{ij}|^{1+\mu}} \theta(|V_{ij}| - N^{-\gamma/2}). \quad (24)$$

where the distribution follows a power-law decay, and is called heavy-tailed for $\mu < 2$ (variance undefined) and fat-tailed for $\mu < 1$ (mean undefined). The typical amplitude of the off-diagonal hopping elements V_{ij} is of order $N^{-\gamma/2}$. Following similar graphical methods as in the Gaussian case, one can show that the scaling exponent c of the typical value for $\Gamma \sim N^{-c}$ is given by $c = \gamma - c - 2(1 - c)/\mu$, where we note the restricted interval for parameters to be $2/\mu < \gamma < 2$ and $1 < \mu < 2$ for the (multi)fractal case [3]. Therefore, we arrive at

$$c = \frac{\mu\gamma - 2}{2(\mu - 1)} \quad (25)$$

One can define an effective parameter γ_{eff} given by $c = \gamma_{eff} - 1$ (same formalism as Gaussian case, as an effective hopping strength parameter), which means

$$\gamma_{eff} = \frac{(2 + \gamma)\mu - 4}{2(\mu - 1)} \quad (26)$$

For the fractal dimension D_q , we have

$$D_q = 1 - c = 2 - \gamma_{eff} = \frac{\mu(2 - \gamma)}{2(\mu - 1)} \quad (27)$$

for all integer q where $q > 1/2$. And it is thus clear that $D_1 = D_2 = \dots = D_\infty$, meaning the Lévy case is also simply fractal but not multifractal, even though the spectrum is indeed different due to heavy tails.

It is also possible to identify from (27) that $\gamma_{ET} = 2/\mu$ and $\gamma_{AT} = 2$ correspond to the ergodic and Anderson localization limits respectively in the interested interval $1 < \mu < 2$.

2 Application to Quantum Algorithms

The initiatives of this section are as follows. First, for the narrow range search problem, we wish to find the detailed prescription for the classical Hamiltonian and understand why fractal (or multifractal) states are fundamentally important for the proposed algorithm in [5]. Second, we identify the **Random Energy Model (REM)** and also stress the relevance of NEE states there [6].

2.1 Narrow Range Search Problem

Recently, there is a proposal for a **Population Transfer (PT)** protocol based on coherent quantum evolution under a constant transverse field rather than adiabatic annealing or Grover search, aiming to exploit many-body tunneling through non-ergodic delocalized states [5]. It is emphasized that an intermediate regime exists where eigenstates are extended but not ergodic (this "NEE" nature is important), forming minibands (instead of the entire spectrum) over which quantum transport occurs efficiently which can be tailored to a specific subclass of optimization problems, and the difference between (mono)fractal and multifractal states is not the key.

2.1.1 Specifying the Problem and Hamiltonian

Quite different from usual unstructured Grover search or optimization problems, the problem to be solved is as follows: given one low-energy bit string (a deep local minimum), one wishes to find other bit strings with nearly the same low energy but are far away in Hamming distance. Concisely, the input is a known low-energy marked state z_j , and the goal is to identify other marked states $z \neq z_j$ whose energy lies within a narrow target window $\Delta\mathcal{E}_{cl}$ around the initial z_j . The most general form of classical Hamiltonian model H_{cl} assumed in [5] is in the form

$$H_{cl} = \sum_z \mathcal{E}(z) |z\rangle \langle z| \quad (28)$$

where $\mathcal{E}(z)$ is any classical energy function over 2^n bit strings $z = (z_1, z_2, \dots, z_n)$ with $z_k = \{0, 1\}$. Here the main qualitative assumptions are:

- (i) The energy landscape of $\mathcal{E}_{cl}$ has many local minima.
- (ii) These low-energy minima are separated by substantial Hamming distances (at least) $d \sim O(n)$.
- (iii) The number of relevant low-energy states is exponential in n yet exponentially small compared to 2^n (which makes it computationally difficult).

Actually, a more specific classical Hamiltonian (partly inspired by the impurity band model in doped semiconductors) is described in [5] which allows more detailed analysis:

$$H_{cl} = \sum_{j=1}^M \mathcal{E}(z_j) |z_j\rangle \langle z_j| \quad (29)$$

for $M \gg 1$, and which is evolved under a "driver Hamiltonian" (responsible for the PT protocol)

given by

$$H_D = -B_\perp \sum_{k=0}^n \sigma_x^k \quad (30)$$

Together we get

$$H = H_{cl} + H_D \quad (31)$$

In (29), it is further assumed that:

- (i) Only the marked states have low energy, and all the other states have $\mathcal{E} = 0$. These unmarked states are also separated by an enormous energy gap $\simeq n$ from the narrow band of marked states.
- (ii) Marked states $|z_j\rangle$ are chosen uniformly and randomly from all bit strings of length n (thus, from all 2^n possible strings). And there is no preferred internal structure or correlation between any of them (except maybe the Hamming distance requirement)
- (iii) Marked states have energies $\mathcal{E}(z_j)$ fall independently around $-n$ in a narrow band of width $W \ll B_\perp$, so are approximately given by

$$\mathcal{E}(z_j) \simeq -n + \epsilon_j \quad (32)$$

where $|\epsilon_j| \ll W/2$.

2.1.2 Introduction to Methods and Results

The general methodology presented in [5] is as follows. One first projects the full spin Hamiltonian onto the marked-state subspace (downfolding), then computes the resulting long-range tunneling amplitudes using WKB theory. Then it is possible to treat the resulting effective Hamiltonian as a random **Preferred Basis Lévy Matrix (PBLM)**, analyze its spectral properties using both exact diagonalization and a nonlinear cavity method, and use the Born approximation to develop intuition for the population-transfer dynamics. Finally, an estimate of the runtime is also performed. Full detail is too much to be summarized here, but a few important notes are included below.

In the model (31), considering the existence of multiqubit tunneling, an eigenstate $|\psi_\beta\rangle$ of H associated with the impurity band can become delocalized over a large subset of marked states $\mathcal{S}_\beta$ with size

$$1 \ll |\mathcal{S}_\beta| \propto M^\alpha, \quad 0 < \alpha < 1. \quad (33)$$

For $\alpha = 0$ the eigenstate $|\psi_\beta\rangle$ is localized, for $\alpha = 1$ the eigenstate is delocalized in the entire space of marked states. For $0 < \alpha < 1$ the eigenstate can be considered “non-ergodic” and its support set $\mathcal{S}_\beta$ is sparse in the space of the marked states.

While the original Hamiltonian H ($2^n \times 2^n$) is sparse in the basis of states, as it couples only states separated by Hamming distance ~ 1 , the down-folded Hamiltonian $\mathcal{H}$ is a dense $M \times M$ matrix. The key difference of the ensemble of matrices $\mathcal{H}_{ij}$ from standard Lévy matrices is that the dispersion, W , of the diagonal matrix elements is much larger than the typical magnitude of the off-diagonal elements V_{typ} , where $V_{typ} \sim n^2 2^{-n/2} e^{-n/(4B_\perp^2)}$.

Generally, comparing the sizes of width W with V_{typ} , we again have the three regimes

$$\begin{cases} W \lesssim V_{\text{typ}} M^{1/2}, & \text{Ergodic,} \\ V_{\text{typ}} M^{1/2} \ll W \ll V_{\text{typ}} M, & \text{Non-ergodic delocalized (NEE),} \\ W \gtrsim V_{\text{typ}} M, & \text{Localized.} \end{cases} \quad (34)$$

where it is emphasized that only the NEE phase is crucial for PT.

A quadratic speed-up to the best classical algorithms ($O(2^n)$) is expected, as quantum PT finds another state within a target window of energies Ω in time $t_{\text{PT}} \propto 2^{n/2} \Omega^{-1/2} \exp(n/(2B_{\perp}^2))$. The scaling exponent of t_{PT} with n differs from that in Grover's algorithm by a factor proportional to $B_{\perp}^{-2}$, which can be made small with large transverse fields $n \gg B_{\perp}^2 \gg 1$. This is similar in efficiency to Grover's algorithm in quantum computing, but crucial distinctions must be noted: non-integrability in this model, and the delocalized nature of the eigenstates within the energy band W .

2.1.3 Importance of Fractality

This section is mainly devoted to concisely answer two questions: First, why is fractality important? Why use the regime of NEE states instead of localized or ergodic ones? Second, is it really necessary to exhibit multifractality (non-degenerate D_q)? Is it sufficient to stay fractal only?

To answer the first question, it is perhaps more clearly explained by ruling out the localized and fully ergodic phases completely. Referring to (33), if one has $\alpha = 0$, the system is completely localized with just each eigenstate confined to itself, without possibility of population transfer, as the wavefunction does not spread to other marked states at all. For the fully ergodic case, $\alpha = 1$, so the eigenstates extend over the entire space of the marked states, and transport would no longer be selective. More importantly, the entire concept of "miniband" will break down, which is fundamental for the specific problem (to identify other marked states whose energy lies within a narrow target window $\Delta\mathcal{E}_{cl}$).

Therefore, we conclude that the requirements of delocalization and non-ergodicity are both crucial for the PT protocol in the narrow search optimization problem.

As for whether genuine multifractality is important, it appears there is no assumption for it in executing the PT algorithm in [5]. For example, key quantities such as the runtime, the miniband width, or the number of states in a miniband are all independent of whether there are multiple fractal dimensions available. It is possibly true, however, that the detailed statistics and dynamics of the NEE states in use would differ according to whether multifractality is present or not.

Even if one has $D_q = D$ for our NEE phase (just fractal), as long as one satisfies that $0 < D < 1$ for the requirement of being delocalized and non-ergodic, all the properties required for PT would appear to be satisfied, such as the formation of minibands and possibility of well-controlled coherent transport. The mechanism is still valid, and this also shows why the non-multifractal Rosenzweig-Porter model can be utilized.

2.2 Random Energy Model

Here we mainly discuss the properties and dynamics of NEE states in the Random Energy Model (REM) [6].

The classical REM introduced contains energies $\mathcal{E}$ which follow a zero-mean normal distribution with standard deviation $\sigma\sqrt{n}$. Its full prescription in the Hamiltonian formalism also follows (31) with the addition of a transverse Hamiltonian. A detailed analysis of properties finds that there are two thermodynamic phases separated by the transition line $B_{\perp}(\beta)$ where β is the inverse temperature.

A naive argument might assume that for $B > B_{\perp}$, the initial state corresponding to a particular spin configuration with low-lying energy $\mathcal{E} \sim O(n)$ should quickly decay into the eigenstates of the system. However, it might not be the case, as [6] points out that, if

$$\sqrt{n} \gg B_{\perp} \gg \frac{1}{\sqrt{n}} \quad (35)$$

then such states can survive for exponentially long times because the spectrum separates into alternating bands, which is the remarkable intermittency feature in the low-energy part of the eigenspectrum of H .

There are two types of bands in the spectrum:

- (I) x-type, which originates from the transverse-field Hamiltonian. Their eigenstates have well-defined x-magnetization, and each band behaves approximately like a GOE random matrix.
- (II) z-type, which are from the classical low-energy minima outside of the vicinity of the transverse field effects. They include MBL states and NEE minibands responsible for long-lived quantum dynamics and population transfer, crucial properties to be shown.

The main body then discusses the calculation detail for these NEE states in similar fashion as [5], i.e. downfolding into a dense Hamiltonian, the use of PBLM, etc.

One important result follows from the case of sufficiently large coupling parameter γ , and the low-energy eigenstates are no longer many-body localized but instead organize into non-ergodic extended minibands. These minibands contain exponentially many computational basis states while still occupying only an exponentially small fraction of the Hilbert space. Their fractal dimension D_q is calculated to be

$$D(|e|, B_{\perp}) = 1 - \frac{2e^2}{\sigma \ln 2} - \frac{2\theta(B_{\perp}/e)}{\ln 2}. \quad (36)$$

where $e = \mathcal{E}/n$ is the energy density, $B_{\perp}$ is the transverse field strength, and σ is the REM disorder strength. $\theta(x) \simeq 1/(4x^2)$ is the approximated WKB tunneling exponent function. Therefore, one can see that the fractal dimension depends on both the energy density (temperature) and the transverse field.

Then one can investigate the detail of the population transfer mechanism. Instead of spreading over the whole Hilbert space, the wavefunction of NEE states remains inside one NEE miniband, and population transfers only among the basis states belonging to that miniband. Such dynamics

is quantitatively explored by studying the statistics of the **Edwards-Anderson (EA)** parameter q_{EA} . Whilst $q_{EA} \simeq 1$ originally as spins remain where they were, whilst for $t \sim 1/\Gamma$ (where Γ is the typical width), then $q_{EA} \sim 2^{-nD}$ tends to decrease exponentially and reveal the fractal dimension D . The long-time memory of the initial spin configuration is controlled directly by the fractal dimension.

Other properties such as the definition of the mobility edge, equipartition of decay rates and "resonances" of nearest neighbours are discussed in [6]. But most crucially there is **asymptotic orthogonality** in the model. Asymptotic orthogonality means that even for very large transverse fields $B_\perp \gg 1$, the cumulative weight of low-energy eigenstates over the typical off-resonant states is exponentially small $\sim e^{-ne^2/2B_\perp^2}$. This situation prevents substantial mixing of the low-energy support sets and leads to the longevity of the z and x states for basically the entire range of $B_\perp$, thereby protecting the low-energy NEE minibands from thermalizing into the ergodic bulk.

The existence of NEE states in the spin-glass model is possibly useful for quantum machine learning and other computational advantages, but such are simply mentioned in [6] which mostly focuses on analyzing the above dynamics.

3 Other Quantum Algorithms

3.1 Non-Hermitian Quantum Adiabatic Algorithm

This section focuses on the theories and methods raised by [7]. Particularly, the consequences of pseudospectral instability arising from conventional non-unitary quantum circuits and **Feynman-Kitaev (FK)** constructions are discussed, and a novel method of **history-decoupled (HD)** mapping is introduced where it has a real, gapped spectrum and is pseudospectrally stable.

3.1.1 Motivations

To understand why non-Hermitian **quantum adiabatic algorithms** (QAA) are needed, we must first explore the central idea of QAA: to encode a given problem into the ground state of a target Hamiltonian, which is adiabatically evolved along a path, during which the energy gap between the ground state and the first excited state must remain finite, to allow desired result at the end following only the ground state evolution (no mixing).

The basic Hamiltonian is laid out as

$$H(s) = (1 - s)H_{init} + sH_{final} \quad (37)$$

where $H(0) = H_{init}$ is the initial Hamiltonian to be prepared and $H(1) = H_{final}$ is the final target Hamiltonian as the solution to the given problem. One can then define a gap $\Delta(s)$ between the ground state and the first excited state as

$$\Delta(s) = E_1(s) - E_0(s) \quad (38)$$

which should be finite throughout the evolution, so that the ground state is slowly evolved to the desired solution.

There is also a direct mapping between quantum circuit (QC) and QAA, so an arbitrary unitary QC process can be encoded as the ground state of a local, Hermitian propagation Hamiltonian.

The framework of unitary QC and Hermitian QAA is well-understood, but there have been recent interests in building non-Hermitian QAAs or non-unitary QCs which are often straightforward to design as they offer extra degrees of freedom. However, there are two issues arising: the complex spectrum of non-Hermitian QAA and the stability of the pseudospectrum. The first one can be easily resolved whereas the second one is the main focus of the text.

3.1.2 FK Construction and Pseudospectral Instability

The simple idea demonstrated in (37) and (38) is often hard to realize in difficult problems, where the gap $\Delta(s)$ may be exponentially small and lead to exponentially long evolution times. Even if these are avoided, finding such optimized Hamiltonian paths are non-trivial. However, one can rely on the **Feynman-Kitaev (FK)** construction to ensure a valid QAA path. This construction maps a L -step unitary QC in the form of $|\psi_L\rangle = U_L U_{L-1} \dots U_2 U_1 |\psi_0\rangle$ into a time dependent Hamiltonian path in the QAA formalism:

$$H(s) = sH_{FK} + (1-s)H_{init} \quad (39)$$

where

$$H_{FK} = \sum_{l=1}^L h_l \quad (40)$$

with

$$h_l = \frac{1}{2} (I_w \otimes |l-1\rangle\langle l-1| + I_w \otimes |l\rangle\langle l| - U_l \otimes |l\rangle\langle l-1| - U_l^{-1} \otimes |l-1\rangle\langle l|) \quad (41)$$

It is also possible to show that after successive adiabatic evolution, one obtains the final ground state: (the initial state is $|\Psi_0\rangle = |\psi_0\rangle \otimes |0\rangle$)

$$|\Psi_L\rangle = \frac{1}{\sqrt{L+1}} \sum_{l=0}^L W_l |\psi_0\rangle \otimes |l\rangle \quad (42)$$

where $W_l = U_l U_{l-1} \dots U_2 U_1$ corresponds to the cumulative circuit after the l -th step.

The important point here is that the minimum energy gap $\Delta \sim \Theta(L^{-2})$ only decreases polynomially with step size L , not exponentially small, ensuring a feasible Hamiltonian path for Hermitian QAAs or unitary QCs.

If such schemes are to be extended to non-Hermitian QAAs, then one must require them to possess a real spectrum which contains an unambiguous definition of an energy gap isolating the ground state from the first excited state. The most straightforward necessary and sufficient condition for the existence of real spectra in non-Hermitian Hamiltonians is the possibility to be transformed

to a Hermitian Hamiltonian H_0 via a similarity transformation given by:

$$H = SH_0S^{-1} \quad (43)$$

where $S = \sum_n |r_n\rangle \langle n|$ is an invertible similarity transform operator with $S^{-1} = \sum_n |n\rangle \langle l_n|$ ($\{|n\rangle\}$ is an orthonormal basis and $|r_n\rangle, \langle l_n|$ are the right and left eigenstates of H). H_0 and H then share an identical and real spectrum.

The other problem is tricky. It is not guaranteed that the eigenvalues within our real spectra remain stable under small perturbations. The fundamental issue is that a non-Hermitian Hamiltonian H is generally non-normal, and hence the non-unitary transformation S can stretch (or compress) and tilt the orthogonal eigenstates of the Hermitian Hamiltonian H_0 which may compromise the orthogonality. This ruins the linearly independent components of the eigenstates which distinguish themselves in the context of small perturbations, causing possible mixture of eigenvalues/eigenstates and an unstable spectrum. This implies the failure of the QAA.

A quantitative tool to characterize the stability (or sensitivity) to perturbations is the ϵ -pseudospectrum defined as:

$$\Lambda_\epsilon(H) = \{z \in \mathbb{C} : \|(zI - H)^{-1}\| \geq \epsilon^{-1}\} \quad (44)$$

with the threshold energy $\epsilon > 0$ associated with the Hamiltonian H .

Graphically, this is the complex region spanned by the eigenvalues under perturbations. Particularly a gap with isolated pseudospectra from the ground state and the first excited state is stable, whilst a fully merged one from the two states is considered unstable, as schematically shown in [7]. The FK QAA's effective model contains accumulation of the non-normality of the similarity transformation, leading to an unstable pseudospectrum (the latter "merged" type).

3.1.3 Solution: HD QAA Construction

The failure of the FK construction in its pseudospectral instability is related to its simultaneous encoding of the entire history of the non-unitary QC. There are multiple ways to explain this, and one way is to see that the similarity transformation $S = \sum_l W_l \otimes |l\rangle \langle l|$ leads to an estimate of the instantaneous energy spectral gap $\Delta(s)$ which is dominated by W_l , composed of many accumulated singularities from multiple non-unitary gates exponentially.

Therefore, one needs to avoid or minimize the influence of such accumulation of pseudospectral instability and consider an alternative method. One way proposed by the method is the **history-decoupled (HD)** QAA: here the Hamiltonian at any instant is dependent only on the current non-unitary operator V_l . whilst the entire history of the quantum evolution is only encoded in the quantum state.

One can parametrize the HD QAA as follows:

$$H(t) = H_l(s(t)), \quad H_l(s) = \Omega(I - P_l(s)) \quad (45)$$

where Ω is the parameter for the gap, and $s(t)$ is a function of time whose detail is not too relevant to be explicitly shown here. $P_l(s)$ is a projection operator in a dynamical bi-orthogonal basis of the composite space of both the work space $\mathcal{H}_w$ and the auxiliary clock space $\mathcal{H}_c = \text{span}\{|0\rangle, |1\rangle, \dots, |L\rangle\}$. For the l -th segment, one obtains that

$$H_l(1) = H_{l+1}(0) = \Omega (I - I_w \otimes |l\rangle\langle l|) \quad (46)$$

which demonstrates a Hamiltonian path that is both continuous and smooth. It is possible to show the pseudospectra of the decoupled Hamiltonians $H_{\sigma_j}(s)$ as the area enclosed by a given finite boundary [7]. The pseudospectrum is controlled locally and does not accumulate into the entire QC.

Physically, whilst the FK QAA encodes the complete QC at once into a static global Hamiltonian H_{FK} in (40), the HD QAA Hamiltonian only links the l -th layer of the QC, coupling only the neighboring $|l-1\rangle$ and $|l\rangle$ sites. It is thus "history-decoupled". As only one clock transition is active at any instant, the non-accumulative nature of non-Hermiticity implies the existence of a stable pseudospectrum, in contrast to the FK construction discussed above.

Moreover, one notes that the subsequent HD Hamiltonian in (46) has a constant gap Ω instead of the $\Theta(L^{-2})$ polynomially decaying gap in the FK construction as in (42). This significantly decreases the required evolution time to achieve our adiabatic algorithm.

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